

Step 1 – Output

```
$ ps -u bread -o pid,comm,stat
PID COMMAND      STAT
519  bash           Ss
522  bash           Ss
527  bash           Ss+
535  bash           Ss+
570  systemd        Ss
571  (sd-pam)       S
591  pipewire        Ssl
592  pipewire        Ssl
594  wireplumber     Ssl
595  pipewire-pulse  Ssl
597  bash            S+
606  dbus-daemon     Ss
1984 nano           S+
4135 gvfsd          Ssl
4214 xdg-desktop-por Ssl
4226 xdg-document-po Ssl
4231 xdg-permission- Ssl
4242 xdg-desktop-por Ssl
4254 at-spi-bus-laun Ssl
4265 dbus-daemon     S
4268 at-spi2-registr Sl
6503 ps             R+
```

Step 2 – Output

```
$ ps -u bread -o pid,comm,stat
PID COMMAND      STAT
519  bash           Ss
522  bash           Ss
527  bash           Ss
535  bash           Ss+
570  systemd        Ss
571  (sd-pam)       S
591  pipewire        Ssl
592  pipewire        Ssl
594  wireplumber     Ssl
595  pipewire-pulse  Ssl
597  bash            S+
606  dbus-daemon     Ss
1984 nano           S+
4135 gvfsd          Ssl
4214 xdg-desktop-por Ssl
4226 xdg-document-po Ssl
4231 xdg-permission- Ssl
4242 xdg-desktop-por Ssl
4254 at-spi-bus-laun Ssl
4265 dbus-daemon     S
4268 at-spi2-registr Sl
6467 forkSample.exe S+
6468 ls             Z+
6481 ps             R+
```

- We can “ls” is the zombie process. It has the Z+ stat next to it.

Step 3 – Output

```
$ ./forkSample.exe
Child's Parent process ID (PPID): 6572
Parent process ID: 6572
$ child's Parent process ID (PPID): 524
```

- The original parent was terminated while the child was sleeping. When that happened, the child was orphaned. I assume that process 524 is used for child process clean-up once they complete execution?